STAT 581: M-estimation and Maximum Likelihood

Notes 4: Handwritten MLE Derivations and Lecture Complements

Liang-Cheng Chen

Updated November 2, 2026

Sources and editorial conventions

This volume integrates all fifteen substantive pages of the updated `../notes/stat581-note4.pdf`; page 16 is blank. Pages 1–5 retain the October 28 material; pages 6–15 add the continuation supplied November 2 (the handwritten date on p. 6 reads November 1). Sections on likelihood, consistency, smooth asymptotics, and sandwich inference incorporate that handwriting. Later sections retain the previously prepared complements from Yen-Chi Chen's *Lecture 3: M-estimation and the maximum likelihood estimator*, pp. 3-1–3-26, including Appendices A–B. The influence-function, QMD/LAN, nuisance-parameter, and optimization sections remain lecture supplements beyond this handwritten source. Gray boxes explain marginal questions, red boxes show corrections, and blue boxes supply additional arguments. Yellow boxes preserve and answer the four explicit TODOs on handwritten pp. 7, 9, 10, and 15. Handwritten M1–M4 labels below are local labels and differ from the numbering in lecture Theorem 3.1. All mathematics uses equation or align environments.

Contents

1 Likelihood, population targets, and misspecification **3**

2 Consistency before normality **4**

3 Classical smooth asymptotic normality **5**

3.1 The handwritten proof, step by step 6

3.2 Remainders, intermediate points, and integrability 7

3.3 Sandwich inference and correct specification 9

3.4 Empirical bootstrap for inference 10

4 Examples of empirical risk minimization **11**

4.1 Least squares, basis expansions, and learned features 11

4.2 Classification and logistic loss 12

4.3 Kernel density derivatives and mode bandwidths 13

5 Modern M-estimation: nonsmooth empirical criteria **14**

5.1 Sample median 15

5.2 Nonlinear least squares 15

6 Influence functions and deletion diagnostics **16**

7 Quadratic mean differentiability and LAN **16**

8 Many nuisance parameters: the missing-outcome example **18**

9 Computational learning: gradient descent **18**

10 Transfer population curvature to the empirical objective **19**

10.1 Computational error versus statistical error 20

1 Likelihood, population targets, and misspecification

Handwritten p. 1; Lecture 3, Sections 3.1–3.2, pp. 3-1–3-7. For IID data under their true law, define the working model and criterion:

$$L_n(\theta) = \prod_{i=1}^{n} p_\theta(X_i), \quad \ell_\theta(x) = \log p_\theta(x), \tag{1}$$

$$M_n(\theta) = \frac{1}{n}\sum_i \ell_\theta(X_i), \quad M(\theta) = \mathbb{E}_P \ell_\theta(X), \tag{2}$$

$$\hat{\theta}_n \in \operatorname*{argmax}_\theta M_n(\theta), \quad \theta^* \in \operatorname*{argmax}_\theta M(\theta). \tag{3}$$

The expectation is under the true law, even if the model is incorrect. When the relevant KL divergences are defined, maximizing expected log-likelihood minimizes the KL divergence from the true law to the working model. With a correct identifiable model the target is the true parameter. Pointwise LLN alone does not prove convergence of maximizers. An interior differentiable maximizer satisfies the score equation; a boundary maximizer or an unbounded likelihood need not do so.

Empirical risk minimization uses the negative criterion. A Z-estimator instead approximately solves an estimating equation. A stationary point need not be a global maximizer.

$$R_n(\theta) = \frac{1}{n}\sum_i L(\theta, X_i), \quad R(\theta) = \mathbb{E}L(\theta, X), \tag{4}$$

$$S_n(\theta) = \frac{1}{n}\sum_i s_\theta(X_i), \quad s_\theta(x) = \nabla_\theta \ell_\theta(x), \quad S(\theta) = \mathbb{E}s_\theta(X), \tag{5}$$

$$H_n(\theta) = \frac{1}{n}\sum_i \nabla_\theta^2 \ell_\theta(X_i), \quad H(\theta) = \mathbb{E}\nabla_\theta^2 \ell_\theta(X). \tag{6}$$

Interchanging derivatives and expectation requires its own domination or integrability justification.

Randomness and the population MLE (handwritten pp. 1–2)

The working density and the true data law need not coincide. The population criterion and its unique maximizer are deterministic functions of the true law and the chosen model; the empirical criterion and its maximizer depend on the data. Thus the pseudo-true target may change when the working model changes, even though the true law stays fixed.

Correction: A pointwise SLLN is not a uniform law (handwritten p. 1)

For each fixed parameter, integrability gives the usual SLLN. Its quantifier does not allow the parameter to vary with the sample or justify exchanging limit and maximization. A uniform law for the log-likelihood class, together with separation

and approximate maximization, supplies the missing argument. A standard sufficient setting is compact parameter space, almost-sure continuity in the parameter, and an integrable envelope for the absolute log-likelihood. The classical empirical-CDF Glivenko–Cantelli theorem alone is not a uniform law for an arbitrary likelihood class.

Two meanings of score (handwritten p. 2)

Here the score differentiates the log-likelihood with respect to the model parameter. In some machine-learning contexts, "score" instead means a gradient with respect to the data. These are different objects:

$$s_\theta(x) = \nabla_\theta \log p_\theta(x), \qquad s_\theta^{\text{data}}(x) = \nabla_x \log p_\theta(x). \tag{7}$$

The latter requires a differentiable continuous-data model. An interior smooth optimum satisfies a score equation, but a score root alone need not be the desired maximizer.

2 Consistency before normality

Theorems 3.7–3.8, pp. 3-8–3-9. A useful argmax theorem assumes uniform convergence, a separated maximum, and sufficiently accurate empirical optimization:

$$\sup_{\theta\in\Theta} |M_n(\theta) - M(\theta)| = o_P(1), \tag{8}$$

$$\forall \varepsilon > 0 : \quad \sup_{\|\theta-\theta^*\|\geq\varepsilon} M(\theta) < M(\theta^*), \tag{9}$$

$$M_n(\tilde{\theta}_n) \geq M_n(\theta^*) - o_P(1). \tag{10}$$

Then the approximate maximizer is consistent. To prove it, call the strict population gap outside the target neighborhood a positive number. Outside that neighborhood, two uniform errors and the optimization error would have to exceed the gap. The probability of this tends to zero. Compactness, continuity, and a unique maximum are common sufficient conditions for separation; uniqueness alone on an arbitrary noncompact space is insufficient.

The Z-estimator analogue replaces maximum separation by a norm bound:

$$\sup_\theta \|S_n(\theta) - S(\theta)\| = o_P(1), \quad S(\theta^*) = 0, \tag{11}$$

$$\inf_{\|\theta-\theta^*\|\geq\varepsilon} \|S(\theta)\| > 0 \quad \text{for every } \varepsilon > 0, \tag{12}$$

$$\|S_n(\tilde{\theta}_n)\| = o_P(1). \tag{13}$$

Uniform convergence and the approximate-zero condition preclude a root far from the unique separated population zero. In vector problems the norms are essential; a vector cannot be ordered by a scalar greater-than sign.

3 Classical smooth asymptotic normality

Theorem 3.1 and Remark 3.2, pp. 3-3–3-5. Use an interior, identifiable target; consistency from the preceding argmax theorem; a score equation; an invertible limiting Hessian; a finite score second moment; and local uniform Hessian convergence. An integrable absolute envelope on third derivatives is one convenient way to control Hessian changes. Assume the log-likelihood and its lower derivatives have the needed integrability at an anchor point.

Correction: The listed M1–M4 need supporting conditions

Compactness and a third-derivative envelope alone do not imply finite score variance or integrability of lower derivatives. A random linear term has zero third derivative but may have infinite variance. The proof also needs consistency before its Taylor remainder is negligible. We state these requirements explicitly rather than infer them from M1–M4 alone.

Set the Hessian and score covariance at the target as

$$H = H(\theta^*), \quad B = \mathbb{E}[s_{\theta^*}(X)s_{\theta^*}(X)^\mathsf{T}], \quad \mathbb{E}s_{\theta^*}(X) = 0. \tag{14}$$

A multivariate-safe expansion integrates the Hessian along a segment:

$$0 = S_n(\theta^*) + A_n(\hat{\theta}_n - \theta^*), \tag{15}$$

$$A_n = \int_0^1 H_n(\theta^* + t(\hat{\theta}_n - \theta^*))\, dt \xrightarrow{P} H, \tag{16}$$

$$\sqrt{n}(\hat{\theta}_n - \theta^*) = -H^{-1}\frac{1}{\sqrt{n}}\sum_i s_{\theta^*}(X_i) + o_P(1) \tag{17}$$

$$\xrightarrow{d} N_d(0, H^{-1}BH^{-\mathsf{T}}). \tag{18}$$

The local segment is in the parameter space with probability tending to one because the target is interior and the estimator is consistent. This expansion avoids assuming a single intermediate point works for an entire vector score. For symmetric Hessians the rightmost inverse transpose is just the inverse.

Correction: Sign and the Taylor remainder

The score-based influence function is the negative inverse Hessian times the score. Theorem 3.1's final expansion and Theorem 3.9 omit this minus sign. It cancels from covariance but matters in linear representations. A second-order score remainder uses the integral weight corresponding to Taylor's theorem; a schematic unweighted integral is only a bound, not an exact remainder formula. The integrated-Hessian identity above is exact.

3.1 The handwritten proof, step by step

Handwritten pp. 2–5; lecture Theorem 3.1. The handwritten assumptions are: M1, both empirical and population targets solve their score equations; M2, the population target is interior; M3, entrywise Hessian integrability at the target and nonsingularity of its expectation. To complete the proof, also assume consistency, finite score second moment, and local uniform control of the Hessian, as stated above. These local handwritten labels are not the lecture's M1–M4 list.

Step 1: fix the target before applying the CLT. Define the empirical fluctuation at the deterministic population target:

$$Y_i = s_{\theta^*}(X_i), \qquad \mathbb{E}Y_i = 0, \tag{19}$$

$$\Delta_n = S_n(\theta^*) - S(\theta^*) = \frac{1}{n}\sum_{i=1}^{n}(Y_i - \mathbb{E}Y_i), \tag{20}$$

$$B = \mathrm{Cov}(Y_i) = \mathbb{E}Y_iY_i^{\mathsf{T}}, \qquad \sqrt{n}\Delta_n \xrightarrow{d} N_d(0, B). \tag{21}$$

At the fixed target these are IID vectors. Evaluating every summand at the same data-dependent estimator generally makes them dependent, so a direct IID CLT cannot be applied to those plug-in terms. The empirical score at the fitted estimator is in fact zero under M1.

Step 2: use the score equations and Taylor expansion. Write the estimation error and expand about the population target:

$$h_n = \hat{\theta}_n - \theta^*, \tag{22}$$

$$-\Delta_n = S_n(\hat{\theta}_n) - S_n(\theta^*) = H_n(\theta^*)h_n + R_n, \tag{23}$$

$$h_n = -H_n(\theta^*)^{-1}\Delta_n - H_n(\theta^*)^{-1}R_n. \tag{24}$$

Correction: Track the remainder sign (handwritten pp. 3–4)

If the remainder is added in the preceding Taylor expansion, it is subtracted after solving for the estimation error. Its sign may be redefined, but both displayed equalities must then use that convention. It cannot simply be set to zero. The integrated-Hessian argument above justifies its negligibility and works for a vector score.

For example, an integrable local envelope for absolute third derivatives bounds the remainder by a bounded-in-probability multiple of the squared error norm. Consistency first makes this remainder small relative to the error norm; the score CLT and nonsingular curvature then yield the root-sample-size rate. Only after that step is the remainder negligible at the CLT scale:

$$\|R_n\| \le C_n\|h_n\|^2, \qquad C_n = O_P(1), \tag{25}$$

$$h_n = o_P(1) \implies R_n = o_P(\|h_n\|), \tag{26}$$

$$\|h_n\| = O_P(n^{-1/2}), \qquad \sqrt{n}R_n = o_P(1). \tag{27}$$

The rate follows by absorbing the small multiple of the error norm into the left side of the solved Taylor inequality. It is not an assumption that can be silently used to prove itself.

Step 3: average IID Hessian matrices and invert the limit. For every coordinate pair assume the indicated absolute integrability:

$$W_i = \nabla^2_\theta \ell_{\theta^*}(X_i), \qquad \mathbb{E}|(W_i)_{jk}| < \infty, \tag{28}$$

$$H_n(\theta^*) = \frac{1}{n}\sum_i W_i \xrightarrow{P} \mathbb{E}W_i = H, \tag{29}$$

$$\det H \neq 0 \quad \Longrightarrow \quad H_n(\theta^*)^{-1} \xrightarrow{P} H^{-1}. \tag{30}$$

The last assertion holds on an event with probability tending to one. The absolute value belongs inside the expectation in the LLN assumption. In fixed dimension, nonsingularity means every eigenvalue is nonzero; for a smooth interior maximum the symmetric Hessian is negative semidefinite, and nonsingularity then makes it negative definite.

Step 4: apply Slutsky's theorem. The resulting covariance and influence function are

$$\sqrt{n}h_n = -H^{-1}\sqrt{n}\Delta_n + o_P(1) \xrightarrow{d} N_d(0, \Sigma), \tag{31}$$

$$\Sigma = H^{-1}BH^{-\mathsf{T}}, \qquad \mathrm{IF}(x) = -H^{-1}s_{\theta^*}(x). \tag{32}$$

The inverse transpose is required by the general covariance transformation; for a symmetric Hessian it equals the ordinary inverse.

3.2 Remainders, intermediate points, and integrability

Handwritten pp. 8–10; Lecture 3, Remark 3.2. The handwritten M4 bounds absolute third derivatives on a neighborhood of the target by an integrable envelope. For a convex local neighborhood, write this as

$$\sup_{\theta\in U}\max_{j,k,l} |\partial_j\partial_k\partial_l \ell_\theta(x)| \le \Lambda(x), \qquad \mathbb{E}\Lambda(X) < \infty. \tag{33}$$

In fixed dimension this gives the earlier remainder bound with a constant times the sample average of the envelope. An exact componentwise formula is

$$R_{n,j} = \int_0^1 (1-t)\sum_{k,l} \partial_j\partial_k\partial_l M_n(\theta^* + th_n)h_{n,k}h_{n,l}\,dt. \tag{34}$$

The remainder includes two factors of the estimation error; it is not merely a third derivative. Absolute values are essential in the envelope.

TODO: Can second derivatives suffice in several dimensions? (p. 9)

Yes. The warning about a common intermediate point is correct, but the conclusion that third derivatives are necessary is not. Applying the scalar mean-value theorem along the parameter segment separately to each score coordinate gives

$$S_{n,j}(\hat{\theta}_n) - S_{n,j}(\theta^*) = H_{n,j\cdot}(\tilde{\theta}_{n,j})h_n, \tag{35}$$

$$\tilde{\theta}_{n,j} = \theta^* + t_{n,j}h_n, \qquad 0 < t_{n,j} < 1. \tag{36}$$

These points generally differ across rows. Stack the rows into a matrix; it need not be a Hessian evaluated at any single point, or be symmetric. Local uniform Hessian convergence and continuity of the population Hessian still make that matrix converge to the target Hessian. Equivalently, use the integrated-Hessian identity already given. Twice continuously differentiable log-likelihoods with a local integrable Hessian envelope provide a useful sufficient setting, together with the other normality assumptions. Consistency of the intermediate points alone does not justify a pointwise LLN at those random points; uniform control is needed.

Supplement: Why a common multivariate mean-value point can fail

Even a gradient field need not have such a point. Consider a smooth function and its gradient along the horizontal unit segment:

$$f(x,y) = x^3/3 + x^3y, \qquad F(x,y) = \nabla f(x,y), \tag{37}$$

$$F(1,0) - F(0,0) = (1,1)^\mathsf{T}, \qquad DF(t,0)(1,0)^\mathsf{T} = (2t, 3t^2)^\mathsf{T}. \tag{38}$$

The first coordinate requires one-half, while the second requires the reciprocal square root of three. No single point satisfies both equations. The integral identity remains valid.

Interior and compactness play different roles (p. 9)

An interior target and consistency put the relevant Taylor segment in a small open ball with probability tending to one. Global compactness is one convenient device for consistency and uniform laws; it is not needed for a local Taylor expansion. Compact parameter spaces also have boundaries. If the target is on a boundary, score equations and ordinary Gaussian limits can fail regardless of compactness.

TODO: Can compactness and M4 replace Hessian integrability? (p. 10)

No, not by themselves. A derivative envelope controls changes from an anchor, rather than the random size of the Hessian at that anchor. On a convex bounded region one obtains

$$|\partial_{jk}\ell_\theta(X)| \le |\partial_{jk}\ell_{\theta_a}(X)| + C_d \operatorname{diam}(\Theta)\Lambda(X). \tag{39}$$

Thus one integrable Hessian anchor plus M4 implies an integrable Hessian envelope. Compactness supplies a finite diameter, but supplies no moment condition on that anchor. For a proper likelihood counterexample, let observed covariates be positive with infinite mean and, conditionally on such a covariate, use a normal response with mean equal to the parameter and variance the reciprocal of the covariate. Up to parameter-free terms,

$$\ell_\theta(x,y) = -\tfrac{x}{2}(y-\theta)^2 + c(x,y), \qquad \partial_\theta^2 \ell_\theta = -x, \qquad \partial_\theta^3 \ell_\theta = 0. \tag{40}$$

M4 holds with a zero envelope on a compact interval, but the expected absolute Hessian is infinite. Moreover, integrability and nonsingularity are separate requirements: a derivative envelope cannot ensure invertible curvature or identification. The earlier M3 cannot simply be dropped.

3.3 Sandwich inference and correct specification

A consistent plug-in covariance for the scaled estimator is

$$\widehat{H} = H_n(\hat{\theta}_n), \quad \widehat{B} = \frac{1}{n}\sum_i s_{\hat{\theta}_n}(X_i) s_{\hat{\theta}_n}(X_i)^\mathsf{T}, \tag{41}$$

$$\widehat{\Sigma} = \widehat{H}^{-1}\widehat{B}\widehat{H}^{-\mathsf{T}}, \tag{42}$$

$$\hat{\theta}_{n,j} \;\pm\; z_{1-\alpha/2}\sqrt{\widehat{\Sigma}_{jj}/n}. \tag{43}$$

Consistency of these plug-ins needs suitable local uniform laws and continuity, not merely convergence of the estimator. Under a correct regular model, the information identity yields

$$B = I(\theta_0) = -H, \qquad \Sigma = I(\theta_0)^{-1}. \tag{44}$$

Under misspecification, the outer-product matrix is the score variability under the true law and generally differs from negative expected curvature. Bootstrap replicate covariance estimates the covariance of the estimator itself; multiply it by the sample size to compare it with the asymptotic covariance of the root-sample-size scaled estimator. Bootstrap validity also requires regularity; it is not universal for every M-estimator.

Correction: Information identity has a minus sign (handwritten p. 5)

Under correct specification and differentiation-under-the-integral conditions, differentiation of the unit total mass gives

$$\begin{aligned} 0 &= \nabla_\theta^2 \int p_\theta(x)\,dx && (45)\\ &= \mathbb{E}_\theta\left[\nabla_\theta^2 \ell_\theta(X) + s_\theta(X) s_\theta(X)^\mathsf{T}\right], && (46)\\ I(\theta_0) &= \mathbb{E}_{\theta_0} s_{\theta_0}(X) s_{\theta_0}(X)^\mathsf{T} = -\mathbb{E}_{\theta_0} \nabla_\theta^2 \ell_{\theta_0}(X) = -H, && (47)\\ \Sigma &= I(\theta_0)^{-1} = -H^{-1}. && (48) \end{aligned}$$

The positive score outer-product expectation is not the expected log-likelihood Hessian. The handwriting misses this sign in its correct-model remark. With misspecification, the outer product under the true law is a score-variability matrix; calling it Fisher information must not be taken to imply the information identity.

Which covariance does the sandwich estimate? (handwritten p. 5)

The displayed sandwich estimates the covariance of the limiting scaled estimator. Divide it by sample size for an estimated covariance of the unscaled fitted parameter. Plug-in Hessians and outer products require local uniform convergence, not just a pointwise LLN at a fixed parameter. In a misspecified model the sandwich is needed even when an inverse-Hessian formula is available computationally.

3.4 Empirical bootstrap for inference

Handwritten pp. 6–7; Lecture 3, Remark 3.2. Conditional on the observations, draw a sample of the original size with replacement from the empirical distribution, then refit the same criterion:

$$X_1^*, \dots, X_n^* \overset{\text{IID}}{\sim} P_n, \tag{49}$$

$$\hat{\theta}_n^* \in \operatorname*{argmax}_{\theta \in \Theta} \frac{1}{n} \sum_i \ell_\theta(X_i^*). \tag{50}$$

Repeat independently for a chosen number of replicates. This avoids explicit analytic score and Hessian evaluation, but requires repeated optimization. In a regular fixed-dimensional problem the conditional law of the centered bootstrap error consistently approximates the sampling law:

$$\mathcal{L}^*\{\sqrt{n}(\hat{\theta}_n^* - \hat{\theta}_n)\} \implies_P N_d(0, \Sigma). \tag{51}$$

Correct model specification is not required for a regular empirical bootstrap: resampling approximates the true data distribution, so it can recover the sandwich limit at the pseudo-true target. Moment consistency is additionally needed when estimating covariance from that limit. With independent replicates and sufficiently many of them, define

$$\bar{\theta}^* = \frac{1}{B} \sum_{b=1}^{B} \hat{\theta}_n^{*(b)}, \tag{52}$$

$$C_B^* = \frac{1}{B-1} \sum_{b=1}^{B} (\hat{\theta}_n^{*(b)} - \bar{\theta}^*)(\hat{\theta}_n^{*(b)} - \bar{\theta}^*)^\mathsf{T}. \tag{53}$$

This estimates the covariance of the unscaled estimator. The estimator of the asymptotic covariance is sample size times this matrix. A fixed number of replicates retains Monte Carlo error. A normal interval uses the square root of a diagonal entry of this matrix directly. Basic bootstrap intervals instead subtract conditional error quantiles from the fitted parameter; tests require an appropriate centered or studentized reference distribution.

Correction: Bootstrap covariance has a sample-size factor (p. 7)

The handwriting calls the raw replicate covariance an estimate of the scaled limiting covariance. It estimates that covariance divided by sample size. This distinction also applies to the lecture's informal bootstrap covariance description.

TODO: Bootstrap and nonconvex parameter spaces (p. 7)

Nonconvexity alone does not imply bootstrap failure. If a unique interior target is separated from other components and refitting finds the correct optimum, the local problem can still be regular. Conversely, failure can occur in a convex space: estimate a nonnegative normal mean at a true mean of zero with unit observation variance. Then

$$\hat{\theta}_n = \max(0, \overline{X}_n), \qquad \sqrt{n}\hat{\theta}_n \xrightarrow{d} \max(0, Z), \qquad Z \sim N(0,1). \tag{54}$$

The centered ordinary bootstrap has a random limiting shift because it refits near a random distance from the boundary. It need not reproduce the fixed target law. Here the ordinary unconstrained sandwich formula also fails to describe the constrained estimator: its normality assumptions are violated. Other failure mechanisms include multiple competing maxima, nonsmooth functionals, and unstable numerical refits. Neither method is automatically protected by the shape of the parameter space.

4 Examples of empirical risk minimization

Handwritten pp. 11–13; Lecture 3, Examples 3.3–3.5. For a loss, empirical risk minimization targets the population minimizer:

$$\hat{\theta}_n \in \arg\min_{\theta} R_n(\theta), \qquad \theta^* \in \arg\min_{\theta} R(\theta). \tag{55}$$

MLE is the special case of negative log-likelihood. For a smooth loss, replace negative log-likelihood curvature by positive risk curvature:

$$A = \mathbb{E}\nabla_\theta^2 L(\theta^*, X), \quad V = \mathrm{Cov}\{\nabla_\theta L(\theta^*, X)\}, \tag{56}$$

$$\sqrt{n}(\hat{\theta}_n - \theta^*) \xrightarrow{d} N_d(0, A^{-1}VA^{-\mathsf{T}}). \tag{57}$$

This needs the same consistency, moment, invertibility, and local regularity conditions; being an ERM alone does not imply asymptotic normality.

4.1 Least squares, basis expansions, and learned features

Squared prediction loss and its empirical risk are

$$L(\theta; x, y) = (m_\theta(x) - y)^2, \qquad R_n(\theta) = \frac{1}{n}\sum_i (m_\theta(X_i) - Y_i)^2. \tag{58}$$

A constant first feature supplies an intercept. For a scalar covariate, polynomials or other fixed basis functions give

$$m_\theta(x) = \sum_{j=1}^{q} \theta_j \phi_j(x), \qquad \phi_1(x) = 1, \quad \phi_2(x) = x, \quad \phi_3(x) = x^2. \tag{59}$$

An orthonormal basis is another option relative to a specified measure. Although nonlinear in the original covariate, this model is linear in its coefficients. A neural feature extractor followed by a regression head generalizes the same construction:

$$z_0 = x, \quad z_{r+1} = \sigma(W_r z_r + b_r), \tag{60}$$

$$\Phi_\eta(x) = z_L, \qquad m_{\beta,\eta}(x) = \beta^\mathsf{T} \Phi_\eta(x). \tag{61}$$

Is this feature engineering? (question on p. 12)

Yes, in the broad sense of constructing useful inputs for prediction. Fixed polynomials are manually chosen features; an autoencoder or neural network learns a representation. If the representation is trained on the same data, its randomness and dependence must be included in inference. Classical fixed-dimensional M-estimation is applicable only when its regularity assumptions hold for the full fitted problem. Growing neural networks, nonidentifiable weights, or nonsmooth activations require further analysis. A frozen externally trained representation is a simpler setting.

Least squares minimizes average squared residuals. For linear prediction, the population target is a best linear projection, even if the conditional mean is nonlinear:

$$\theta^* = \operatorname*{argmin}_\theta \mathbb{E}(Y - X^\mathsf{T}\theta)^2. \tag{62}$$

4.2 Classification and logistic loss

The handwritten classification extension can use a logistic head on fixed or learned features, with binary responses. For fixed features, the following loss derivatives give the smooth ERM score and curvature.

Logistic regression uses the loss and probability

$$q_\theta(x) = (1 + e^{-x^\mathsf{T}\theta})^{-1}, \tag{63}$$

$$L(\theta; x, y) = -yx^\mathsf{T}\theta + \log(1 + e^{x^\mathsf{T}\theta}), \tag{64}$$

$$\nabla L = x(q_\theta(x) - y), \quad \nabla^2 L = xx^\mathsf{T} q_\theta(x)(1 - q_\theta(x)). \tag{65}$$

A finite unique minimizer is an assumption; complete separation can destroy existence of an unrestricted logistic MLE.

Classification with zero–one loss is also ERM, but the criterion is usually nondifferentiable, so ERM membership alone does not imply the classical smooth normality theorem applies.

4.3 Kernel density derivatives and mode bandwidths

Handwritten pp. 14–15; Lecture 3, Example 3.6. For a kernel density estimate and its mode,

$$\hat{p}_h(x) = \frac{1}{nh^d} \sum_i K((X_i - x)/h), \tag{66}$$

$$\widehat{m}_h \in \operatorname*{argmax}_x \hat{p}_h(x), \quad m_h \in \operatorname*{argmax}_x \mathbb{E}\hat{p}_h(x). \tag{67}$$

For fixed bandwidth this is an M-estimator for the smoothed mode. For a shrinking bandwidth it is a triangular problem: under additional kernel, density, curvature, and bandwidth conditions the scale is

$$\sqrt{nh^{d+2}}(\widehat{m}_h - m_h) \xrightarrow{d} N(0, \Sigma_h^{\lim}). \tag{68}$$

The lecture's statement is a schematic rate. Centering instead at the true mode also requires controlling smoothing bias, often of order squared bandwidth under appropriate smoothness.

For a univariate density and a symmetric probability kernel, use

$$\hat{p}_h(x) = \frac{1}{nh} \sum_i K((X_i - x)/h), \qquad K(u) = \frac{1}{\sqrt{2\pi}} e^{-u^2/2} \quad \text{for the Gaussian kernel.} \tag{69}$$

With sufficient density smoothness and finite kernel second moment, the pointwise bias is of order squared bandwidth. For a square-integrable kernel and a continuous density at the evaluation point,

$$\mathbb{E}\hat{p}_h(x) = p(x) + \tfrac{1}{2}h^2\mu_2(K)p''(x) + o(h^2), \tag{70}$$

$$\operatorname{Var}(\hat{p}_h(x)) = \frac{p(x)}{nh} \int K(u)^2\, du + o((nh)^{-1}), \tag{71}$$

$$\mu_2(K) = \int u^2 K(u)\, du. \tag{72}$$

Correction: Density derivative versus data score (p. 15)

The derivative of a density is not its log-density score:

$$g(x) = p'(x), \qquad s^{\text{data}}(x) = p'(x)/p(x) \quad \text{when } p(x) > 0. \tag{73}$$

The handwritten derivative estimate targets the former. A log-density score estimate would divide by the density estimate and needs control where the density is small.

For a differentiable kernel with square-integrable derivative, and under sufficient smoothness for the bias expansion,

$$\hat{g}_h(x) = -\frac{1}{nh^2} \sum_i K'((X_i - x)/h), \tag{74}$$

$$\mathbb{E}\hat{g}_h(x) = p'(x) + \tfrac{1}{2}h^2\mu_2(K)p'''(x) + o(h^2), \tag{75}$$

$$\mathrm{Var}(\hat{g}_h(x)) = \frac{p(x)}{nh^3}\int K'(u)^2\,du + o((nh^3)^{-1}). \tag{76}$$

Assume a unique separated interior mode, strictly negative second density derivative there, and a consistent local mode estimator. A derivative Taylor expansion and uniform local curvature convergence give

$$\widehat{m}_h - m^* = -\frac{\widehat{p}_h'(m^*)}{p''(m^*)} + o_P((nh^3)^{-1/2}) \tag{77}$$

when the bandwidth makes the derivative bias negligible on that scale.

TODO: What does a proper bandwidth speed mean? (p. 15)

Bandwidth must decrease to remove smoothing bias while leaving enough observations to estimate derivatives. One convenient sufficient rate regime in one dimension is

$$h \to 0, \qquad \frac{nh^5}{\log n} \to \infty, \qquad nh^7 \to 0. \tag{78}$$

Under standard smoothness, bounded-kernel derivative and local uniform empirical-process conditions, the middle requirement ensures local second-derivative consistency; the last removes the first-derivative bias at its CLT scale. These rates supplement, rather than replace, mode separation and consistency assumptions. A power bandwidth has the useful range

$$h = n^{-a}, \qquad 1/7 < a < 1/5. \tag{79}$$

Then a triangular-array derivative CLT and the mode expansion yield

$$\sqrt{nh^3}(\widehat{m}_h - m^*) \xrightarrow{d} N\left(0, \frac{p(m^*)\int K'(u)^2\,du}{[p''(m^*)]^2}\right). \tag{80}$$

Indeed, the scaled bias is of order the square root of sample size times bandwidth to the seventh power. At the usual mode mean-squared-error balance with bandwidth proportional to sample size to power minus one seventh, that bias is generally nonzero in the limit. Centering at the smoothed population mode instead removes this bias term, subject to the corresponding local derivative conditions. Shrinking bandwidth creates a triangular-array problem, so the fixed-bandwidth M-estimator theorem alone cannot establish this result.

5 Modern M-estimation: nonsmooth empirical criteria

Theorem 3.9, Examples 3.10–3.11, pp. 3-9–3-11. A useful normality theorem requires: differentiability at the target for almost every observation; a local Lipschitz envelope with

finite second moment; a nonsingular second-order population expansion; consistency; and an empirical criterion within a term smaller than inverse sample size of the empirical optimum (exact maximizers satisfy this).

$$M(\theta) = M(\theta^*) + \tfrac{1}{2}(\theta - \theta^*)^\top H(\theta - \theta^*) + o(\|\theta - \theta^*\|^2), \tag{81}$$

$$|\ell_{\theta_1}(x) - \ell_{\theta_2}(x)| \le \eta(x)\|\theta_1 - \theta_2\|, \quad \mathbb{E}\eta(X)^2 < \infty. \tag{82}$$

Then the same negative-Hessian score expansion applies. The sample criterion need not have a second derivative everywhere.

5.1 Sample median

Assume a unique population median with a density continuous and positive there. For a finite expected absolute loss, additionally assume a finite first moment (alternatively work with centered loss differences).

$$\ell_\theta(x) = -|x - \theta|, \quad m = F^{-1}(1/2), \tag{83}$$

$$s_m(x) = \operatorname{sign}(x - m) \quad (x \neq m), \tag{84}$$

$$M'(\theta) = 1 - 2F(\theta), \quad H = M''(m) = -2f(m), \tag{85}$$

$$\psi(x) = \frac{\operatorname{sign}(x - m)}{2f(m)}, \tag{86}$$

$$\sqrt{n}(\widehat{m} - m) \xrightarrow{d} N\left(0, \frac{1}{4f(m)^2}\right). \tag{87}$$

The local Lipschitz envelope is one. An observation equal to the target has probability zero, so its kink does not violate almost-everywhere differentiability at that target. A consistent density plug-in or a valid median bootstrap supplies uncertainty estimates.

Correction: Median maximization has negative curvature

Example 3.10 reverses the signs in the absolute-loss integrals, score, and Hessian. For negative absolute loss the population maximum has negative curvature. The displayed limiting variance in the lecture is nevertheless correct because the signs cancel when squared.

5.2 Nonlinear least squares

Suppose the conditional mean model is correct, errors have conditional mean zero, and the needed differentiability and integrability hold:

$$Y = m_{\theta^*}(X) + e, \quad \mathbb{E}[e \mid X] = 0, \quad g(X) = \nabla_\theta m_{\theta^*}(X), \tag{88}$$

$$\ell_\theta(X, Y) = -\tfrac{1}{2}(Y - m_\theta(X))^2, \tag{89}$$

$$s_{\theta^*}(X, Y) = e\, g(X), \quad A = \mathbb{E}[g(X)g(X)^\top], \quad H = -A, \tag{90}$$

$$B = \mathbb{E}[e^2 g(X) g(X)^\mathsf{T}], \quad \Sigma = A^{-1} B A^{-1}. \tag{91}$$

Under constant conditional variance this reduces to sampling variance times the inverse of the Gram matrix. The half in the criterion ensures the displayed derivative has no factor two. Without it both the score and Hessian double, leaving the sandwich unchanged.

Correction: Curvature is not automatically Fisher information

Example 3.11 drops derivative factors and the negative curvature sign. The Gram matrix is not generally Fisher information for an arbitrary error law. A likelihood and error distribution are required to make that identification. The constant population criterion at the truth is negative one-half the error second moment with the convention used here.

6 Influence functions and deletion diagnostics

Section 3.3.3, p. 3-12. For the maximization convention,

$$\hat{\theta}_n - \theta^* = \frac{1}{n} \sum_i \psi(X_i) + o_P(n^{-1/2}), \quad \psi(x) = -H^{-1} s_{\theta^*}(x). \tag{92}$$

Large influence indicates sensitivity. For the corresponding linearized estimators, deleting one observation changes the average by

$$\hat{\theta}_n - \hat{\theta}_{n,-k} \approx \frac{1}{n} \left\{ \psi(X_k) - \frac{1}{n-1} \sum_{i \neq k} \psi(X_i) \right\}. \tag{93}$$

Correction: The deletion effect has an inverse-sample-size factor

Section 3.3.3 omits this factor. The influence function describes a scaled first-order sensitivity. The ordinary asymptotic-linear remainder alone is too coarse to prove an inverse-sample-size deletion approximation; this interpretation requires additional smoothness or a direct deletion analysis. The displayed identity is exact for the linearized averages.

7 Quadratic mean differentiability and LAN

Section 3.4, Theorems 3.13–3.14, pp. 3-12–3-17. QMD is differentiability of the square-root density in the square-integrable function space with respect to a common dominating measure:

$$\int \left[\sqrt{p_{\theta+h}} - \sqrt{p_\theta} - \tfrac{1}{2} h^\mathsf{T} s_\theta \sqrt{p_\theta} \right]^2 d\mu = o(\|h\|^2). \tag{94}$$

The full-support formulation is essential when support changes. A formal pointwise derivative on the interior of the support is insufficient. QMD implies zero-mean square-integrable score under the model. In the normalized Hellinger convention of Lecture 2,

$$H^2(P_{\theta+h}, P_\theta) = \tfrac{1}{8} h^\mathsf{T} I(\theta) h + o(\|h\|^2). \tag{95}$$

If Hellinger squared is defined without the one-half integral factor, the coefficient is one-quarter instead.

For a uniform law on an interval with unknown right endpoint, a positive endpoint shift creates a new support interval whose square-root-density error is already first order in the shift:

$$\int_\theta^{\theta+h} p_{\theta+h}(x)\, dx = \frac{h}{\theta + h}, \qquad h > 0. \tag{96}$$

Hence the model is not QMD at the usual root-sample-size scale. Indeed, the endpoint MLE is the sample maximum and has the nonnormal limit

$$n(\theta - \max_i X_i)/\theta \xrightarrow{d} \mathrm{Exp}(1). \tag{97}$$

This last limit is an editorial calculation: use the exact CDF of the maximum.

For a correct QMD model, with bounded local vectors tending to a fixed vector, the LAN expansion under the target model is

$$\log \prod_i \frac{p_{\theta+h_n/\sqrt{n}}(X_i)}{p_\theta(X_i)} = h_n^\mathsf{T} \Delta_n - \tfrac{1}{2} h_n^\mathsf{T} I(\theta) h_n + o_P(1), \tag{98}$$

$$\Delta_n = \frac{1}{\sqrt{n}} \sum_i s_\theta(X_i) \xrightarrow{d} N(0, I(\theta)). \tag{99}$$

For a consistent MLE with nonsingular information and the local log-likelihood Lipschitz condition of Theorem 3.14 (with an envelope having a finite second moment),

$$\sqrt{n}(\hat{\theta}_n - \theta_0) = I(\theta_0)^{-1} \Delta_n + o_P(1) \xrightarrow{d} N(0, I(\theta_0)^{-1}). \tag{100}$$

Correction: Hellinger, KL, and proof qualifications

The source switches Hellinger normalizations. Fixing the convention avoids a factor-two error. Its equation (3.11) equates the expected log-likelihood difference with minus twice an unnormalized Hellinger square; that is only a local second-order relation when an expected log-likelihood expansion is justified, not an exact identity. QMD controls square-root densities; passing a pointwise logarithm remainder through expectation requires more control. Use the LAN theorem and the full assumptions of Theorem 3.14, rather than treating that informal proof as valid from QMD alone.

8 Many nuisance parameters: the missing-outcome example

Section 3.5, pp. 3-17–3-18. Let there be many categories, with known observation probabilities bounded away from zero and one:

$$X \sim \mathrm{Unif}\{1, \dots, B\}, \quad R \mid X = b \sim \mathrm{Bernoulli}(\xi_b), \tag{101}$$

$$Y \mid X = b \sim \mathrm{Bernoulli}(\theta_b), \quad R \perp Y \mid X, \quad \delta \le \xi_b \le 1 - \delta, \tag{102}$$

$$\psi_B = B^{-1} \sum_{b=1}^{B} \theta_b. \tag{103}$$

Only outcomes with indicator one are observed. In categories with observed outcomes, the MLE is the empirical success proportion; in empty categories the likelihood is flat. With far more categories than observations, most coordinates are unidentified by the observed likelihood and an arbitrary plug-in completion or independent prior can dominate the result.

An inverse-probability estimator targets the functional directly:

$$Z_i = R_i Y_i / \xi_{X_i}, \quad \widehat{\psi}_B = n^{-1} \sum_i Z_i, \tag{104}$$

$$\mathbb{E} Z_i = B^{-1} \sum_b \theta_b = \psi_B, \quad \mathbb{E} Z_i^2 = B^{-1} \sum_b \theta_b / \xi_b \le 1/\delta, \tag{105}$$

$$\mathrm{Var}(\widehat{\psi}_B) \le 1/(n\delta). \tag{106}$$

Chebyshev proves consistency even along growing-category sequences. A triangular-array CLT also gives asymptotic normality if the row variances converge to a positive limit; bounded summands verify Lindeberg. Without variance convergence there need not be a single fixed normal limit.

Correction: Interpret the failure claim in its asymptotic regime

With fixed finitely many categories and positive observation probabilities, every category is eventually observed and standard consistency can return. The difficulty is the growing or very large category regime, not a universal failure of every Bayesian prior or penalized estimator. A penalized MLE is a posterior mode for a matching penalty, not generally the posterior-mean Bayes rule. Estimating the average can succeed without recovering every coordinate.

9 Computational learning: gradient descent

Appendix A, Theorem 3.17, pp. 3-19–3-21. On a convex region suppose a differentiable objective has a Lipschitz gradient and a positive strong-convexity constant:

$$f(y) \le f(x) + \nabla f(x)^{\mathsf{T}}(y - x) + \tfrac{L}{2}\|y - x\|^2, \tag{107}$$

$$f(y) \geq f(x) + \nabla f(x)^{\mathsf{T}}(y - x) + \tfrac{M}{2}\|y - x\|^2. \tag{108}$$

When twice differentiable, these correspond to all Hessian eigenvalues between the two constants. Gradient descent updates

$$\theta^{(t+1)} = \theta^{(t)} - \gamma \nabla f(\theta^{(t)}). \tag{109}$$

For a stationary minimizer and a fixed step smaller than both reciprocal constants, the proof in the lecture yields

$$\|\theta^{(t)} - \hat{\theta}_n\|^2 \leq (1 - \gamma M)^t \|\theta^{(0)} - \hat{\theta}_n\|^2. \tag{110}$$

The argument expands the squared update, uses strong convexity to control the cross term, and smoothness to bound squared gradient by twice the smoothness constant times excess objective value. The remaining gradient term is nonpositive for the stated step size.

10 Transfer population curvature to the empirical objective

Appendix B, Lemmas 3.18–3.19 and Theorem 3.20, pp. 3-22–3-26. For the negative population criterion, suppose the target Hessian has smallest eigenvalue bounded below by a positive constant and is locally Lipschitz in operator norm:

$$\lambda_{\min}(\nabla^2 R(\theta^*)) = \lambda_* > 0, \quad \|\nabla^2 R(\theta) - \nabla^2 R(\theta^*)\| \leq \phi \|\theta - \theta^*\|. \tag{111}$$

Choose a ball lying in the interior of the parameter space, of radius at most the smallest eigenvalue divided by twice the Lipschitz constant. Weyl's eigenvalue inequality then ensures curvature at least half the target curvature throughout that ball.

A compact Lipschitz-indexed class has a uniform LLN when its envelope of increments is integrable and one anchor function is integrable:

$$|f_{\theta_1}(x) - f_{\theta_2}(x)| \leq q(x)\|\theta_1 - \theta_2\|, \quad \mathbb{E}q(X) < \infty, \quad \mathbb{E}|f_{\theta_a}(X)| < \infty. \tag{112}$$

A finite parameter net, ordinary LLN at its finitely many points, and a Lipschitz approximation bound prove this. Applying it to Hessian entries in fixed dimension yields uniform convergence in operator norm.

On an event with probability tending to one, require all of the following: the empirical estimator is within half the population ball radius of the target; the uniform Hessian error is at most one-quarter the target curvature; and empirical Hessian norms are bounded by twice a finite population bound. Then in a ball around the empirical estimator,

$$M^* = \lambda_*/4, \qquad L^* = 2h_{\max}, \qquad \|\theta^{(t)} - \hat{\theta}_n\| \leq (1 - \gamma M^*)^{t/2} \|\theta^{(0)} - \hat{\theta}_n\|. \tag{113}$$

The contraction keeps iterates in the local ball if initialization lies there. There is no guarantee that an arbitrary initial point reaches this ball. The lecture's last proof uses the maximum-eigenvalue subscript in several lower bounds where the minimum-eigenvalue subscript is needed; the union bound must count each of its three events once.

10.1 Computational error versus statistical error

The two errors add by the triangle inequality:

$$\|\theta^{(t)} - \theta^*\| \leq \|\theta^{(t)} - \hat{\theta}_n\| + \|\hat{\theta}_n - \theta^*\|. \tag{114}$$

A fixed contraction factor requires logarithmically many iterations in sample size to reach the statistical root-sample-size scale. To preserve the estimator's limiting distribution, require computational error to be smaller in probability than that scale, not merely bounded by it. This last comparison is an editorial synthesis of the two lecture appendices.